Ideation Report

Milestone 2

When Is Help Actually Helpful?

Rethinking Studying with AI

Group 3

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A.0 Declaration of Use of AI

We used AI in this report in the following ways.

- **Finding related work.** We described our needs and key ideas to an AI assistant (Claude, through Claude Code), which searched the web for related papers, products and articles.
- **Organizing and synthesizing reading.** We used the assistant to summarize sources, compare multiple articles and identify the most relevant ones for each need. The paper summaries were based on abstracts and web search results, not full texts, and the product descriptions on documentation and reviews, not hands-on use.
- **Writing Markdown.** We used the assistant to organize our notes and write them as Markdown files, including the structure of this report.
- **Dictation.** Some team members used dictation tools, such as Wispr Flow, to type faster.

The research questions, the choice of baselines, the needs and the conclusions are the team's. The team reviewed and edited the AI-assisted text.

A.1 Field Research Summary

We reviewed 19 sources (7 papers, 10 products and 2 other posts or docs), organized by the five key needs from Milestone 1. Findings from papers are based on abstracts. Product findings come from documentation, reviews and web searches, not hands-on use.

Related work

Source	Type	Need	Key insight
Help Helps, But Only So Much	Paper	1	Coaching students on how to seek help made them use it more deliberately but did not improve learning.
Segmentation of Worked Examples	Paper	1	Pre-segmented examples gave equal learning with less effort, but making students segment them backfired.
Thetawise	Product	1	Hints and step-by-step math help, but the student picks Tutor Mode or Answer Directly.
TeachLM	Paper	2	Current study modes rephrase a problem instead of diagnosing the confusion and barely establish the student's level.
Learning Context framework	Paper	2, 3	AI tutors should keep a model of the learner's cognitive, affective and sociocultural context.
Gemini Guided Learning	Product	2	Asks about goal and level first, plus a diagnostic quiz tied to uploaded class material.
ChatGPT Study Mode	Product	2	Asks about goal and level before teaching.
Claude document upload	Product	3	Course documents can be uploaded as context.
ScrollToStudy	Product	3	Builds a feed of flashcards and quizzes from the student's own material, for passive review.
Rundown thought-partner post	Post	4	An agent interviews the user one question at a time before drafting. Shows people using AI as a partner that guides the discussion.

Source	Type	Need	Key insight
Claude Code learning style	Docs	4	Leaves TODOs so the user makes the decisions.
Turbo AI	Product	baseline	Produces notes, flashcards and quizzes but does not guide students through problems.
Beyond the Benefits	Paper	5	Review of 224 papers on the harms of generative AI in computing education, including cognitive effects and trust.
Evolution of Programmers' Trust	Paper	5	Students' trust in Copilot rose over 10 days. Seeing it be right raised trust and wrong suggestions lowered it.
Trust and Reliance on AI in Education	Paper	5	Higher trust went with worse judgment of when the AI was wrong.
Scite	Product	5	Helps judge how trustworthy or contested a scientific claim is.

Related work in depth

Two papers shaped our design most directly.

Segmentation of Worked Examples (Spanjers, van Gog and van Merriënboer, 2012). This study compared three ways of studying worked examples: shown whole, pre-divided into segments by blank lines, and divided into segments by the students themselves. Pre-segmented examples led to the same learning with less mental effort, while asking students to segment actively added effort without improving performance. A related study by the same group found that segmentation helped learners with low prior knowledge more than those with high prior knowledge. For our concept this supports splitting help into short steps (Need 1), and suggests the tool should do the splitting instead of leaving it to the student. The prior-knowledge result also links Need 1 to Need 2, since the right step size may depend on what the student already knows. The study tested worked examples, not AI chat, so it supports the design direction without proving it.

Evolution of Programmers' Trust in Generative AI Programming Assistants (Shah et al.). This study followed 71 upper-division computer science students using GitHub Copilot, measuring their trust after one hour and again after 10 days. Trust rose on average. Students said it grew when they saw Copilot be correct, understood how it uses the surrounding code as context, and learned some basics of how language models work. Incorrect suggestions were a main reason for distrust, and students concluded that Copilot still needs a competent programmer for some tasks. For our concept (Need 5), it shows that students build trust from what they can see the tool get right or wrong, which supports showing sources and confidence for each answer. The study measured attitudes rather than behavior, involved only students with programming backgrounds, lasted 10 days and concerned code completion, not help with STEM problems in general.

The three baselines are covered in the next part: NotebookLM, Khanmigo and GitHub Copilot.

Baseline apps

NotebookLM. Answers only from the sources the student uploads, with inline citations that open the cited passage, and a Learning Guide mode that asks Socratic questions. Its strength is grounding in the student's own material. Its weaknesses are that the student still has to verify

each answer, it does not check what the student already knows, and it shows no confidence marker. The key insight I get from the product is that it's a good opportunity for students to centralize all materials and have an AI assistant that goes over it, still delegating most of the autonomy to the user.

Khanmigo. A tutor built into Khan Academy, so it has the exercise context built in and guides with hints and questions instead of answers. Its strength is focused, question-led help. Its weaknesses are that it only covers Khan Academy content, a review found it ignored a student's correct work, and it has no source references. This tool offers a great opportunity as a reference on how to use AI as guardrails, given that it understands the material and is able to communicate information. There are still some worries about how much correctness the AI provides to the user when asking questions and whether it should be able to respond fully to a question.

GitHub Copilot. A general coding assistant, included because it is widely used and because one study measured students' trust in it. Its strength is adoption and evidence on how trust forms. Its weaknesses are that it is not built for learning, it has no staged help or knowledge check, and its context is code, not course material. This is different from agentic coding in some sense: it's much more active coding than agentic coding, where you're only sitting. In this instance, you have to still write some guidelines very specifically and see the code changing in front of you.

None of the three checks what the student already knows before helping, and none marks how sure an answer is.

Most related solution

The closest existing solution is NotebookLM. Students upload their own course material, and answers come only from it, with inline citations that open the cited passage. Its Learning Guide mode also guides in a Socratic style instead of answering directly. What it does not do, as far as we could find, is check what the student already knows before helping, stage support from a hint up to a full solution, or mark how sure an answer is. Khanmigo comes closest on guided help and has its course context built in, but only for Khan Academy content. Our concept combines the three: the student's own course material with source markers, an up-front knowledge check, and help that escalates only as the student needs it.

A.2 Brainstorming Summary

How might we statements

Each statement maps to one key need from Milestone 1.

- How might we make AI provide step-by-step solutions for educational prompts? (Need 1)
- How might we find out what a student already understands without making it feel like a boring test? (Need 2)
- How might we ensure that AI answers are based on the course material, and make good use of uploaded course material? (Need 3)
- How might we adapt AI so that it does not only provide the answer but instead supports the student more? (Need 4)
- How might we help students verify whether AI-generated help is trustworthy? (Need 5)

Brainstorming method

The brainstorming method we used was round-robin brainstorming. Initially everyone wrote down their own ideas and after the time was up and everyone felt ready we compared and discussed the ideas. We sorted the ideas to match the "how might we" questions. After one round of round-robin we had already found a decent number of ideas so we did not do a second round.

We chose this brainstorming method because it seemed like the most efficient way to quickly collect a big number of ideas. Everyone could write down all that was in their head without being restricted to the previous person's ideas (as it would have been the case in 6-3-5 brainwriting).

Brainstormed ideas

These are the ideas we came up with, sorted by their corresponding how might we question.

HMW 1: How might we make AI provide step-by-step solutions for educational prompts? (Need 1)

- The Moodle to Markdown tool should outline exercise solutions by steps, and thus go one by one guiding the user, hence following the approach of the course.
- Generate a breakdown of the problem and give small inputs to each step.
- Create a harness that penalizes lengthy responses (user defined).
- Reveal one step at a time. The AI shows step 1, and the student taps "next step" or "I'll try the next one" to continue. Maybe have intermediate questions to go to the next step.
- Force the AI answers to have some specific format. There could be multiple formats, like one for ensuring that the question was understood correctly and one for a deep-dive answer.
- Go through the steps until a solution is reached. The user is guided through the solution steps and can give input.

HMW 2: How might we find out what a student already understands without making it feel like a boring test? (Need 2)

- Generate a summary based on provided material and ask the student to self-rate.
- Have an expand option in a step-by-step answer.
- Start from the student's own attempt. The student pastes their partial work or a one-line explanation, and the AI infers their level from it.
- Have a record of courses taken, performance, and current lecture notes.
- Interview the user in the way Tony Ojeda goes about starting a project.
- Make a fun and exciting test to find out what a student already knows.
- Track learning progress by writing down a user's confidence levels for subtopics of their studies.
- Have a user_profile file that is updated with each chat and contains info about the user's knowledge level about subjects.
- Just test them and see if they can solve the tests for specific subjects or problems.
- Strongly incentivise the student to do a test before going into the topic (a firewall of some sort).

HMW 3: How might we ensure that AI answers are based on the course material, and make good use of uploaded course material? (Need 3)

- Responses need to point to a line, slide or section.
- Upload material and make guardrails to avoid answering outside of the scope.
- A Moodle to Markdown tool that gives the AI the necessary course material without saturating the context window.
- Upload your own documents and the AI references them, so one knows where the response was generated.

HMW 4: How might we adapt AI so that it does not only provide the answer but instead supports the student more? (Need 4)

- Give a hint based on course material, and only give the solution when specifically asked (not as the default).
- Intermediate exercise solving to see if the student comprehends.
- Khan Academy style guardrails.
- Instead of immediately giving the answer, the AI progressively increases its support from a small hint, to a concept explanation, to a similar example and eventually a full solution (a learning ladder).

HMW 5: How might we help students verify whether AI-generated help is trustworthy? (Need 5)

- When gathering outside sources, have a list of good websites and a blacklist of bad websites (user defined?).
- Confidence and source markers. The confidence score is based on our judgment (human judgment). Each claim is labeled as "from your course material", "general knowledge" or "unsure".
- Go back to the Moodle to Markdown file and give a cross-reference on the course material of the produced answers, or some level of source that the user can go and check for themselves.
- Upload your own documents and the AI references them, so one knows where the response was generated. Use Scite if an outside source is used.

A.3 Affinity Diagram Summary

The following image shows the affinity diagram we ended up with. The same board is in Figma.

- **Green** sticky notes represent ideas from our brainstorming session.
- **Red** sticky notes represent findings from our desk research.
- **Yellow** sticky notes are the five key needs from Milestone 1.


Student's context (What they know)...
Memory of Student's Knowledge
Need 2: Help that matches what the student already knows
franco
having a record of courses taken; performance; and current lecture notes
reto
track learning progress by writing down a user's confidence levels for subtopics of their studies
reto
Start from the student's own attempt. The student pastes their partial work or a one-line explanation, and the AI infers their level from it.
reto
Generate a summary based on provided material and ask student to self rate
reto
Research on the expertise reversal effect shows that help that works for a beginner can backfire for someone who already knows more, so instruction should be adapted to the learner's level. https://link.springer.com/article/10.1007/s10648-007-9054-3
Rafael Bragança
There is existing research on creating a holistic view of the learner for an AI model
https://arxiv.org/abs/2512.24362
reto
TeachLM researchers tested Claude Learning Mode, ChatGPT Study Mode and Gemini Guided Learning by hand and found that they spend practically no time finding out the student's level, goal or motivation before explaining. https://arxiv.org/abs/2510.05087
Rafael Bragança
Test the Student
review idea?
Interview the user in the way Tony Ojeda goes about starting a project
reto
Make a fun and exciting test to find out what a student already knows.
reto
When solving an exercise, probe the student with intermediate exercises to check whether so far the student has understood what the problem is asking from them (For example asking a slightly different version of what they just solved).
franco
Students need to understand the core idea well enough to solve similar problems alone
Rafael Bragança
Gemini study notebooks start from the student's goal and uploaded class material, run a diagnostic quiz to find what the student still needs to learn, and update the lessons from the quiz results. https://blog.google/innovation-and-ai/products/gemini-app/gemini-study-notebooks/
Rafael Bragança
Intermediate exercise solving to see if the student comprehends
reto
Students need to work out what they are actually stuck on
Rafael Bragança
Strongly incentivise the student to do a test before going into the topic (firewall of some sort)
reto
Students need room to struggle before help arrives
Rafael Bragança
Making answers verifiable
Trusting the source
Need 5: Knowing when not to trust the AI
franco
Confidence and source markers. Confidence score is based on our judgment (human-judgment). Each claim is labeled as "from your course material", "general knowledge" or "unsure".
reto
There are several existing papers on the behavior on students in regards to trusting their AI tools (especially for coding):
https://arxiv.org/abs/2510.04443
https://arxiv.org/abs/2509.13253
https://arxiv.org/abs/2604.01114
valia
References
Responses need to point to a line or slide or section
reto
feature
Add course context
Moodle to markdown tool that gives the AI the necessary course material without saturating context window
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Go back to the moodle to markdown file and give cross reference on the course material of the produced answers, or some level of source that the user can go and check for themselves
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The moodle to markdown should outline exercise solutions by steps, and thus go one by one guiding the user, hence following the approach of the course
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Upload material and make guardrails to avoid answering outside of the scope
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Students need help that fits the kind of subject and task
Rafael Bragança
Upload your own documents and the AI references them thus one knows from where the response was generated.
reto
NotebookLM (Gemini Notebook) is an existing tool that is really good for uploading your course material and having the AI work on that.
reto
Upload your own documents and the AI references them thus one knows from where the response was generated. Use Scite if an outside source is used
reto
Need 3: Help grounded in the course material
franco
AI Answer Format
Use the Khan academy guardrails to prevent direct answer production
franco
According to Tony Ojeda, to prevent the AI of just providing answers, it is good to perform an interview first to build the context the AI needs to support the user in doing the thinking
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Force the AI answers to have some specific format, there could be multiple formats. like one for ensuring that the question was understood correctly and one for a deep dive answer.
reto
Create a harness that penalizes lenghty responses (user defined)
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Anthropic gave Claude a specific "/output-style learning" mode that gives the user TODO fields to act as an incentive for them to keep being part of the thinking rather than just providing answers.
franco
Step-by-step guidance / support
Reveal one step at a time. The AI shows step 1, and the student taps "next step" or "I'll try the next one" to continue. Maybe have intermediate questions to go to next step
reto
Goes through the steps until solution is reached. User is guided through the solution steps and can give input.
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Instead of immediately giving the answer, the AI progressively increases its support from a small hint, to a concept explanation, to a similar example and eventually a full solution. Learning ladder
reto
Need 1: Short, step-by-step help that goes deeper on request
franco
Segmentation of Worked Examples: Effects on Cognitive Load and Learning
concept of cognitive loading: the total amount of mental effort being used in your working memory at any given time
Between the user and the AI, have another agent acting as a supervisor that stops the AI from just giving the answer directly.
franco
Have expand option in a step by step answer
reto
Constructing a progression of the styles of support the AI should give. It starts from small hints, to then concept explanation, to a similar example and eventually the full solution ("Learning ladder")
franco
Need 4: Support that keeps students doing the thinking
franco

Key themes and insights

Grouping the sticky notes produced seven sections. A few of those sit inside larger groups on the diagram. We did not sort the notes by the how-might-we questions again. References is closely related to trusting the source.

Key sections

- Memory of the student's knowledge
- Testing the student
- Trusting the source
- References
- Adding course context
- AI answer format
- Step-by-step guidance and support

Each team member had four stars to place on the sections the prototypes should focus on. The sections we want to focus on are below.

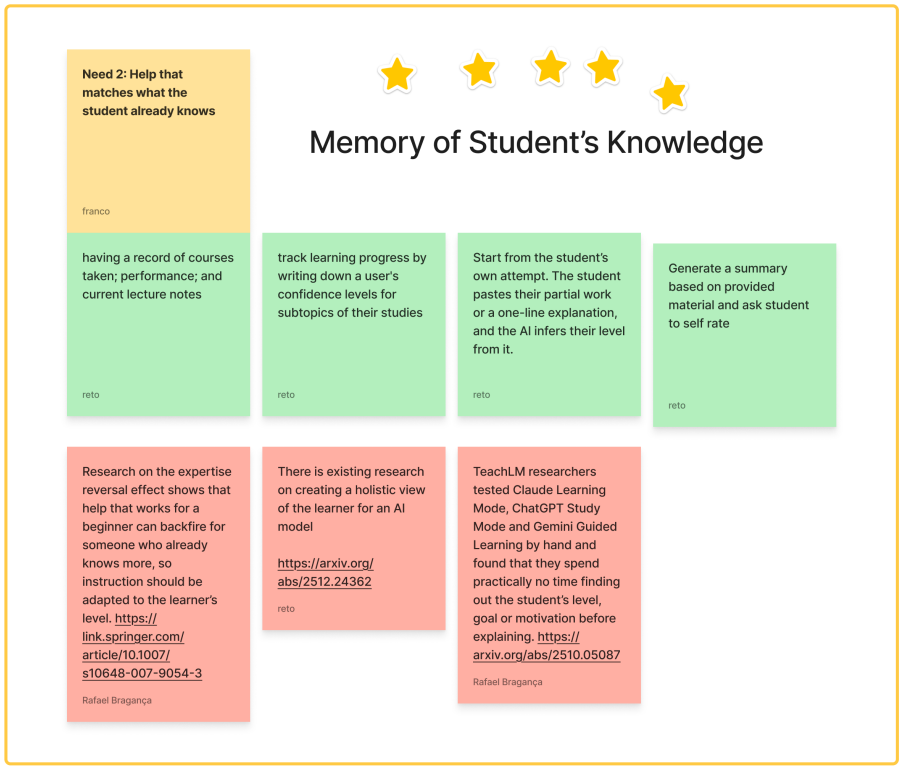
Focus for the prototypes

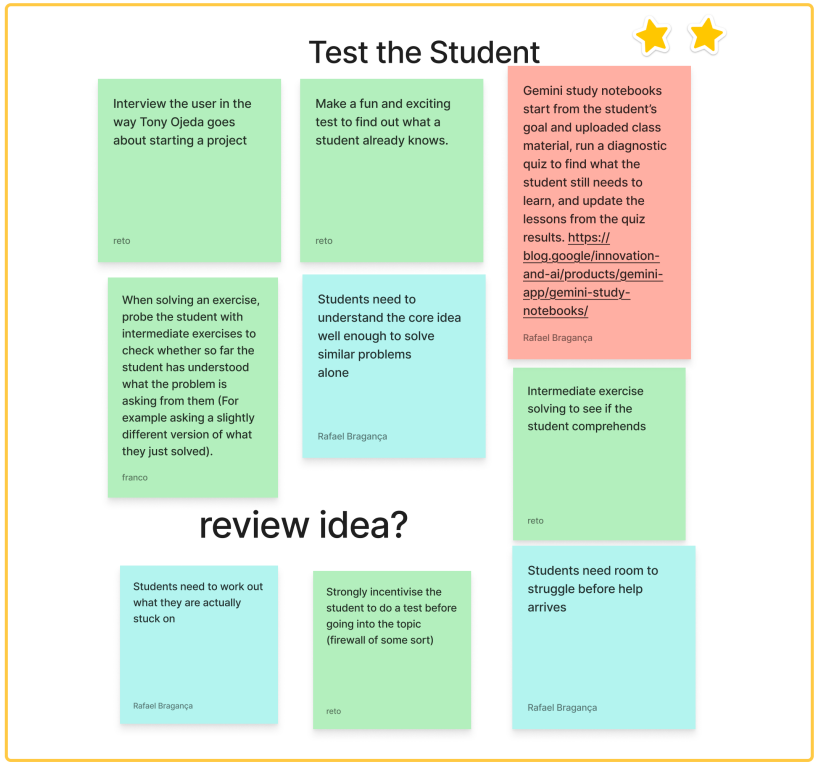
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- Memory of the student's knowledge
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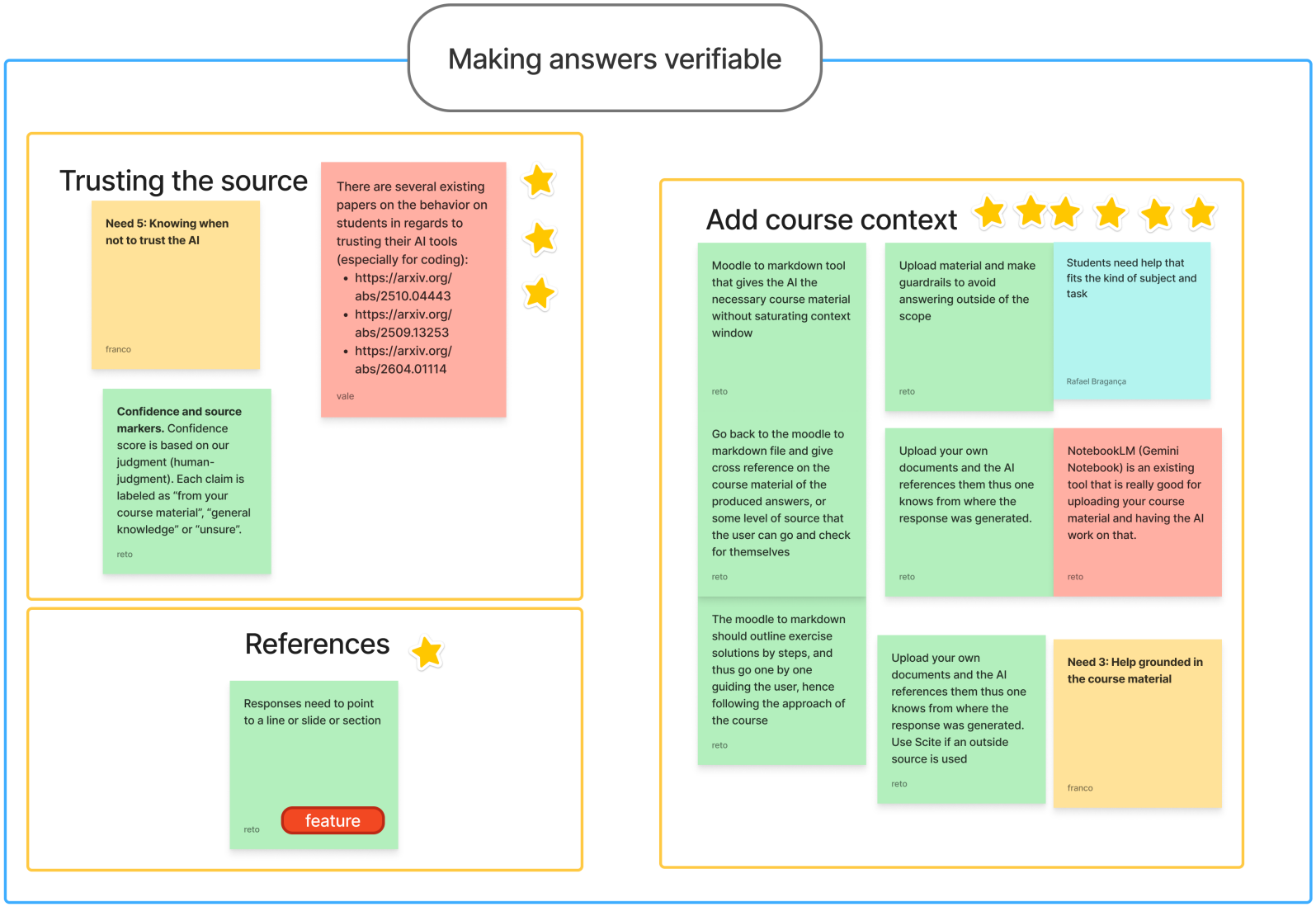
Diagram close-ups

The following pages show each part of the board at a larger size so the sticky notes can be read.

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Trusting the source
Need 5: Knowing when not to trust the AI
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Segmentation of Worked Examples: Effects on Cognitive Load and Learning

concept of cognitive loading: the total amount of mental effort being used in your working memory at any given time

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franco

Need 4: Support that keeps students doing the thinking

franco